

JUAN CARLOS CATTANEO

[PORTFOLIO](#) / [LINKEDIN](#) / [GITHUB](#)

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#PROFESSIONAL SUMMARY

Software Engineer with 5+ years designing hardware-software integration systems and developer-facing APIs for medical devices. Currently leading mobile app development and driving algorithmic innovation. Specialized in full-stack development spanning backend systems (C#, Rust, Python) to mobile frontends (React Native, React), with expertise in cross-platform architecture and high-performance computing.

#TECHNICAL SKILLS

Languages: C#, Python, Rust, C++, JavaScript, TypeScript, Ruby

Frameworks & Libraries: .NET/WPF, React Native, React, Next.js, Flask, FastAPI, Ruby on Rails

Data: MySQL, SQLite, PostgreSQL, MongoDB, Supabase

Cloud + Tools: Azure DevOps, CI/CD, Git, GitHub Actions, OpenCV, Agile / Scrum, NuGet, NPM

IDEs & Development: Xcode, WebStorm, RustRover, Visual Studio, VScode, Rider, PyCharm

Hardware: Cameras, Sensors, Motors, Arduino, Raspberry Pi, Jetson Nano

AI: Claude Code, GitHub Copilot, Ollama, Agentic Development

#PROFESSIONAL EXPERIENCE

Mobile App Team Lead - SWE

June 2026 - Present

Carl Zeiss Meditec

- Mobile App Leadership & Architecture: Leading mobile app development with React Native, architecting cross-platform solutions for medical imaging applications and driving technical direction across iOS and Android platforms.
- High-Performance Algorithm Migration: Translated core algorithmic engine from Python to Rust, 61 files, ~40,400 lines of numerical/scientific computing code, with 0.0001 accuracy threshold. Resulted in significant performance improvements and platform scalability.
- CI/CD Pipeline Development: Architected Azure pipeline infrastructure for automated build, testing, and npm package distribution of algorithm libraries, enabling seamless integration across teams.
- Mobile VR Simulation & Testing: Integrated VR set simulation into React Native mobile app for enhanced device interaction testing and user validation workflows.
- Test Coverage & Quality: Significantly improved React mobile app test coverage from 9% to 92%, fixing approximately 45 significant bugs in the process and establishing foundation for long-term code reliability.

Acquisition IC / Acquisition Team Lead - SWE

Sep 2021 - June 2026

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- Acquisition Team Lead: Leading the acquisition software, driving technical direction for OCT engine development, and coordinating cross-functional efforts across subsystems.
- AI-Assisted Development: Integrated GitHub Copilot and Claude Code into daily workflows to accelerate development, code review, unit tests, and documentation.

- Platform API Development: Designed and maintained an OCT engine interface layer (DLLs/libraries) consumed by multiple teams for medical imaging hardware abstraction.
- API Architecture & Integration: Architected 30+ interface methods for camera, sensor, and motor systems; refactored legacy code while preserving backward compatibility.
- Platform Engineering: Built C# WPF/MVVM solutions with 97%+ unit test coverage and architected CI/CD pipelines for automated build, test, and deployment.
- Complex Systems Debugging: Reduced critical defects by 40% through root cause analysis of multi-subsystem hardware-software integration issues.
- Backend Systems Integration: Built C++ sensor control backend and C# WPF frontend, developed Linux/OpenCV applications for image processing workflows.
- Code Quality & Refactoring: Refactored 80% of legacy WPF codebase using OOP principles, established MS Unit and NUnit testing frameworks.
- Python Automation & Tooling: Built Python APIs and automation tools for hardware validation and data analysis used by systems and research teams.

Software Engineer Intern

June 2021 - Sep 2021

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- Front-End SWE: Built React web app with Hooks. 90%+ unit test coverage.
- SQA: Automated desktop and web tests using Tosca and Testim IO.

#EDUCATION

California State University, East Bay / B.S. Business Administration - IT Management	2019
App Academy Certificate / Full-Stack Software Engineering (1,000+ hrs)	2020
Diablo Valley College / Certificate - Computer Science - Program Design	2023
Diablo Valley College / Associate - Business Administration - Economics	2015